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RESEARCH PAPER

Challenging the Performance-Interpretability Trade-Off: An Evaluation of Interpretable Machine Learning Models

Sven Kruschel · Nico Hambauer · Sven Weinzierl · Sandra Zilker · Mathias Kraus · Patrick Zschech

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Abstract Machine learning is permeating every conceivable domain to promote data-driven decision support. The focus is often on advanced black-box models due to their assumed performance advantages, whereas interpretable models are often associated with inferior predictive qualities. More recently, however, a new generation of generalized additive models (GAMs) has been proposed that offer promising properties for capturing complex, non-linear patterns while remaining fully interpretable. To uncover the merits and limitations of these models, the study examines the predictive performance of seven different GAMs in comparison to seven commonly used machine learning models based on a collection of twenty tabular benchmark datasets. To ensure a fair and robust model comparison, an extensive hyperparameter search combined with cross-validation was performed, resulting in 68,500 model runs. In addition, this study qualitatively examines the visual output of the models to assess their level of interpretability. Based on these results, the paper dispels the misconception that only black-box models can achieve high accuracy by demonstrating that there is no strict trade-off between predictive performance and model interpretability for tabular data. Furthermore, the paper discusses the importance of GAMs as powerful interpretable models for the field of information systems and derives implications for future work from a socio-technical perspective.

Keywords Decision support · Predictive analytics · Interpretable machine learning · Generalized additive models · Explainable artificial intelligence

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S. Kruschel · N. Hambauer · M. Kraus
Chair for Explainable AI in Business Value Creation, University of Regensburg, Bajuwarenstraße 4, 93053 Regensburg, Germany
e-mail: sven.kruschel@ur.de

N. Hambauer
e-mail: nico.hambauer@ur.de

M. Kraus
e-mail: mathias.kraus@ur.de

S. Weinzierl · S. Zilker
Chair of Digital Industrial Service Systems, Friedrich-Alexander-Universität Erlangen-Nürnberg, Fürther Straße 248, 90429 Nuremberg, Germany
e-mail: sven.weinzierl@fau.de

S. Zilker
e-mail: sandra.zilker@th-nuernberg.de

S. Zilker
Professorship for Business Analytics, Technische Hochschule Nürnberg Georg Simon Ohm, Hohfederstraße 40, 90489 Nuremberg, Germany

P. Zschech (✉)
Professorship for Intelligent Information Systems and Processes, Leipzig University, Grimmaische Straße 12, 04109 Leipzig, Germany
e-mail: patrick.zschech@uni-leipzig.de

1 Introduction

Machine learning (ML) has made significant advancements in recent years, allowing for the automation of many tasks related to predictive decision-making (Janiesch et al. 2021). Promising examples can be found in various fields such as e-commerce (Ghavamipoor and Hashemi Golpayegani 2020), transportation (Balster et al. 2020), industrial maintenance (Landwehr et al. 2022; Zschech et al. 2019), and business process monitoring (Oberdorf et al. 2022; Kratsch et al. 2021). However, many advanced ML models, such as boosted decision trees and deep neural networks, suffer from black-box characteristics. This means that the mathematical functions they learn between input features and target values are so complex that it is practically impossible for humans to understand how these models generate a prediction for a given data instance (Bauer et al. 2021). This lack of transparency can lead to a lack of trust in using these models for high-stakes decision tasks, such as in healthcare, finance, and criminal justice (Rudin 2019; Thiebes et al. 2021).

To address the lack of transparency, interpretable ML models have been developed (Barredo Arrieta et al. 2020). The structure of interpretable models is constrained in some way to provide a better understanding of how predictions are generated (Rudin 2019). Traditional representatives are linear models and decision trees, which are easy to analyze and comprehend, but often too limited to capture more complex relationships. As a remedy, more advanced models have been proposed to mitigate the trade-off between predictive performance and interpretability. Of particular interest are *generalized additive models (GAMs)*, which are currently experiencing a renaissance in the debate about comprehensible decision support (Nori et al. 2019; Barredo Arrieta et al. 2020; Zschech et al. 2022).

In general, GAMs are a type of ML model that allows for the estimation of non-linear relationships between predictor variables (i.e., features) and a response variable (i.e., target). They are an extension of linear models, in which the linearity assumption is relaxed, allowing for more flexible and powerful modeling capabilities (Hastie and Tibshirani 1986). More specifically, the relationship between each feature and the target is modeled separately in so-called *shape functions*, and the results are combined in an additive manner. This allows the model to capture arbitrary relationships while remaining fully interpretable, which provides crucial benefits for model analysis and debugging purposes (Lou et al. 2012).

In recent years, multiple versions of GAMs have been proposed. Although they all follow the same principle of using non-linear functions to map input features to the target space, they integrate different algorithmic approaches. For example, while the original GAM uses splines for fitting shape functions (Hastie and Tibshirani 1986; Wood 2017), more recent variants are based on trees (Lou et al. 2012; Caruana et al. 2015) or tailored neural networks (Agarwal et al. 2021; Yang et al. 2021b; Kraus et al. 2024). Taken together, these different extensions show promising directions for increasing model performance while remaining fully transparent. As such, they constitute pivotal instruments in the area of interpretable ML and comprehensible decision support, which is of central importance for the information systems (IS) community.

Despite their promising potential, there have been only a few evaluation studies to empirically assess the merits and limitations of different GAM variants. Each model has already been tested in individual settings and domains, especially by the respective authors and developers. However, it currently lacks an overarching cross-model comparison that considers the different particularities. Likewise, it lacks a thorough analysis that examines the performance gap between these interpretable models and commonly applied black-box models. We believe that such findings are essential to advancing and promoting the use of transparent ML models in research and industry alike.

For this reason, we conduct a comprehensive evaluation study in which we analyze a series of modern GAM variants and compare them with traditional ML models based on a large collection of tabular datasets. In particular, this work contributes to the existing body of knowledge in the following ways:

- We assess and compare the predictive performance of seven different GAMs as well as seven commonly used ML models based on a collection of twenty benchmark datasets.
- We conduct an extensive hyperparameter search and perform a cross-validation to ensure a fair and robust model comparison, resulting in more than 68,500 training runs across all datasets and models.
- We examine the visual outputs of all seven GAMs, verify their intrinsic interpretability, assess similarities and differences between them, and evaluate their level of interpretability based on six evaluation criteria.
- We integrate all models and experiments into a unified evaluation pipeline and provide a publicly available repository that is free to use for researchers and practitioners.
- We summarize our results and dispel the misconception that only black-box models can achieve high accuracy by demonstrating that there is no strict trade-off between predictive performance and model interpretability for tabular data.
- We discuss the importance of GAMs as powerful interpretable models for the IS community and the broader field of decision support by outlining their

overall suitability for high-stakes decision tasks and derive implications for further research from a socio-technical perspective.

The remaining paper is structured as follows: Sect. 2 provides an overview of relevant foundations and related work. Subsequently, Sect. 3 describes the experimental set-up of our evaluation study, followed by the presentation of the results in Sect. 4. Section 5 offers a discussion of our findings, outlines the potentials of advanced GAMs for the IS community, and highlights promising avenues for future research. Section 6 concludes the paper with summarizing thoughts.

2 Conceptual Background and Related Work

2.1 Model Interpretability

The need for model interpretability has grown substantially due to the widespread use of ML models, particularly in critical domains like credit scoring, healthcare, and criminal justice (Rudin and Radin 2019; Bauer et al. 2021). In general, model interpretability refers to a model's ability to explain or present its decision logic in a human-understandable way (Du et al. 2019; Doshi-Velez and Kim 2017). This can take various forms depending on the specific use case and stakeholder requirements. What is enough for one use case may not be enough for another, and different stakeholders may have distinct expectations and requirements for interpretability. To this end, Doshi-Velez and Kim (2017) distinguish between application-grounded, human-grounded, and functionally-grounded interpretability. While the first two involve user-centric evaluation criteria, such as perceived complexity or required cognitive effort, the latter refers to a formal definition of interpretability, allowing for a more objective assessment of interpretability. In this paper, we focus explicitly on the *functionally-grounded* perspective of interpretability and consider the extent to which a model is able to visually reveal how model inputs (i.e., features) are processed in order to produce a predicted outcome (i.e., target).

Apart from that, it is possible to distinguish between post-hoc interpretability and intrinsic interpretability (Rudin 2019; Du et al. 2019). *Post-hoc interpretability* requires the construction of a second model to explain the behavior of an existing model that usually exhibits a high degree of complexity, such as boosted decision trees or deep neural networks. Common post-hoc techniques include Shapley additive explanations (SHAP) (Lundberg et al. 2020) and local interpretable model-agnostic explanations (LIME) (Ribeiro et al. 2016). They are often also subsumed under the term explainable artificial intelligence (XAI) as they provide explanations that attempt to simplify a complex mathematical function so that it is digestible by a human user (Barredo Arrieta et al. 2020). Although post-hoc explanations can lead to valuable insights (e.g., Senoner et al. 2022), they must be considered with caution as they might not be reliable and can result in misleading conclusions (Rudin 2019; Babic et al. 2021).

By contrast, *intrinsic interpretability* is achieved by constructing models that are interpretable by design (Rudin 2019). Such models provide not just an approximate explanation of a model's functional behavior, but an *exact* description of how a model computes a prediction. That is, the underlying mathematical function (which fully defines the model) is simple enough so that users can access and analyze it directly. To achieve this goal, the models are restricted in their structure by incorporating practical constraints, such as linearity, monotonicity, additivity, and sparsity (Rudin and Radin 2019; Sudjianto et al. 2020). Common representatives of intrinsically interpretable models are linear models, point systems, and decision trees, which have often been favored in the past in high-stakes decisions due to their transparency and ease of understanding (Lou et al. 2012; Rudin and Radin 2019; Yang et al. 2021b). At the same time, however, such simple models often lack the flexibility to capture more complex relationships, which naturally occur in real-world applications.

2.2 Generalized Additive Models

In this study, we focus on the specific model family of *GAMs* as a particularly powerful class of intrinsically interpretable ML models that balance transparency with high accuracy. To achieve this goal, GAMs build relationships between input features and the target by summing up several distinct non-linear mappings, known as *shape functions*[1] (Lou et al. 2012). As such, they combine the simplicity of linear models with the flexibility of non-linear mappings by replacing static model coefficients with flexible shape functions to capture more complex relationships between input features and the prediction target.

More formally, this model structure can be expressed as follows. Let $D = (X, y)$ denote a training dataset, where $X = (x_1, \ldots, x_n) \in \mathbb{R}^{N \times n}$ is the feature matrix comprising N samples and n features and y denotes the target variable. For a regression task, y is a real value ($y \in \mathbb{R}^N$), while for a binary classification task, y is a binary value ($y \in \{1, 0\}^N$). A generalized additive model is then defined in the

[1] Alternatively, some works use the term *smooth function* (Hastie and Tibshirani 1986) to emphasize the property that the function is infinitely differentiable and results in a smooth graph.

[2] If the link function is the identity function, Eq. 1 describes a regression model, and if the link function is the logistic function, Eq. 1 describes a classification model.

following form by taking a feature matrix X as input and calculating a prediction $\hat{y}$ as output

$$g(\hat{y}) = f_1(x_1) + \cdots + f_n(x_n), \tag{1}$$

where $g(\cdot)$ is called link function[2] and $f_i(\cdot)$ is the shape function for a feature x_i (Lou et al. 2012).

By employing separate functions for each feature, the overall structure of GAMs remains simply interpretable, as it allows model users and developers to verify the importance of each feature. In other words, it is directly observable how each feature, through its corresponding shape function, affects the predicted output. For the examination of the individual feature effects, the corresponding shape functions can be visualized using two-dimensional shape plots, in which the x-axis represents the feature values and the y-axis represents the impact on the predicted output. On this basis, human experts can evaluate the established relationships and verify them with their domain knowledge.

Figure 1 on the right shows an example of a shape plot where a GAM captures the non-linear relationship between body temperature and the probability (in log odds) of mortality. In contrast, on the left, a linear model is visualized which assumes a linear relationship between body temperature and the probability of mortality. This comparison highlights the flexibility of GAMs in modeling complex patterns, providing more accurate and interpretable predictions as opposed to linear alternatives.

2.3 Variants of Generalized Additive Models

The original GAM refers to a model in which the shape functions represent *regression splines* (Hastie and Tibshirani 1986). Splines are a class of mathematical functions defined over intervals, typically composed of piecewise segments joined at specific points known as knots. They serve as flexible tools for approximating complex relationships in data, employing a variety of functions, such as polynomial functions or radial basis functions, to achieve smoothness and high flexibility in modeling. To address unnecessary model complexity and avoid overfitting, spline-based GAMs can also be enriched with advanced techniques such as shrinkage and automated smoothness estimation techniques to encourage smoother and more parsimonious models (Wood 2017). Moreover, some authors integrate spline-based GAMs into ensemble learning strategies in order to further enhance their predictive performance (e.g., De Bock et al. 2010).

A second stream of research focuses on tree-based learning approaches for fitting shape functions in GAMs (Caruana et al. 2015). To this end, Lou et al. (2012) argue that splines are often too smooth for real-world applications and that higher predictive performance is achievable by using tree-based step functions. More specifically, the authors suggested using bagged and boosted tree ensembles to increase model flexibility and to fit more detailed shape functions. The authors further extended their approach by incorporating pairwise interaction terms, which can still be visualized as two-dimensional heatmaps (Lou et al. 2013). Subsequently, the authors made their algorithm publicly available as an easy-to-use implementation known as *explainable boosting machine (EBM)* (Nori et al. 2019).

Another stream of research has shifted the focus to the development of GAM extensions inspired by tailored artificial neural networks. For example, Agarwal et al. (2021) proposed *neural additive model (NAM)*, in which feature-wise shape functions are learned via deep neural networks consisting of multiple hidden layers. To fit the rapid changes in the target variables of real-world data, the authors introduce exp-centered hidden units. These units allow to capture sharp changes in the output.

Yang et al. (2021b) proposed *generalized additive models with structured interactions (GAMI-Net)*. The basic architecture is similar to that of NAM, using a simple GAM-based structure in the form of a disentangled feed-forward network with multiple additive sub-networks. To reduce unnecessary model complexity and avoid overfitting, GAMI-Net integrates sparsity constraints to select the most relevant features to receive a compact model. Furthermore, GAMI-Net has a similar strength to EBM in that it is able to incorporate pairwise interactions between individual features, which can lead to better predictive performance. At the same time, this functionality requires further model constraints such as heredity and marginal clarity constraints to retain structural interpretability and avoid mutual absorption between main effects and pairwise interactions.

A different architectural design was sought with the proposal of *explainable neural network (xNN)* (Vaughan et al. 2018) and its successor enhanced explainable neural network (ExNN) (Yang et al. 2021a). Similar to GAMI-Net, both models are based on multiple sub-networks and regularization terms to retain sparsity and receive a compact model. However, instead of using a simple GAM structure, both models are based on the structure of additive index models by adopting the idea of a projection pursuit regression (Friedman and Stuetzle 1981). This structure generally violates the principle of univariate feature mappings due to an additional projection layer that fully connects all input features to the following sub-networks. As a result, each feature can possibly have a partial contribution to all corresponding shape functions, to achieve a higher level of predictive performance.

More recently, Kraus et al. (2024) proposed another novel GAM variant, called *interpretable generalized*

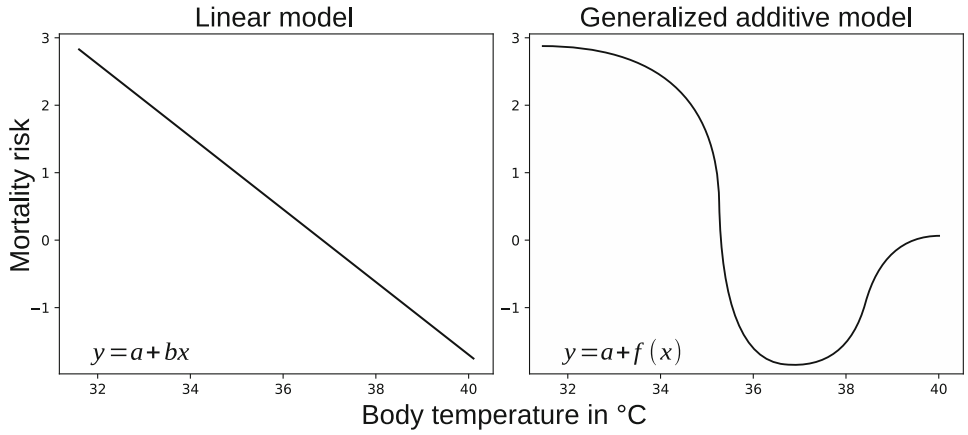

Fig. 1 Comparison of shape functions of a linear model and a GAM

additive neural networks (IGANN). In the first step, the model initializes shape functions linearly and then incrementally adapts to potential non-linearities. That is, IGANN is based on the concept of linear modeling and only resorts to non-linearities if the underlying data requires it. For the incremental adaptation to non-linear shapes, IGANN incorporates the principle of gradient boosting. Specifically, it uses a boosted ensemble of tailored sparsified neural networks, where each network represents an extreme learning machine and acts as a weak learner to gradually improve the performance.

2.4 Comparative Evaluation Studies

All of the GAM-based models presented above offer innovative algorithmic concepts and components to balance model accuracy and transparency. While the main principle remains almost identical for all models, some of the approaches differ greatly in their incorporated learning principles and model constraints. Therefore, it is worthwhile to evaluate and compare their merits and limitations for different prediction tasks and datasets.

A few authors have conducted experiments to compare their proposed GAM extensions to traditional ML baselines and competing models. For instance, Lou et al. (2012) compared spline-based models with tree-based GAMs as well as linear/logistic regression (LR) and random forest (RF) as lower and upper bound baselines. Similarly, Yang et al. (2021b) evaluated GAMI-Net against several benchmark models, including EBM, splines, LR, RF, extreme gradient boosting (XGB), and multi-layer perceptron (MLP), using a large number of datasets. Agarwal et al. (2021) benchmarked NAM against EBM and traditional approaches, albeit with a limited number of datasets.

Beyond that, only a few studies have assessed the characteristics of different intrinsically interpretable models so far. For instance, Chang et al. (2021) examined a series of GAMs, such as spline-based models like penalized cubic regression splines and tree-based models like EBM, using benchmark datasets from real-world applications and simulations. They evaluated the models both qualitatively and quantitatively. Hohman et al. (2019) incorporated GAMs into a visual analytics tool to examine how data scientists interact with shape functions. Meanwhile, Kaur et al. (2020) explored how data science experts use and evaluate interpretable models, including both intrinsically and post-hoc interpretable models like SHAP to compare the results of both paradigms.

In summary, when considering the focus of related work, it currently lacks a cross-model comparison to evaluate the merits and limitations of different GAMs from a broader and more neutral perspective. Likewise, it lacks a thorough analysis that examines the performance gap between these advanced interpretable models and commonly applied black-box models. Especially for application-oriented sciences such as the IS community, it is crucial to understand how different interpretable models behave in direct comparison as well as how they perform in comparison to commonly used ML models. We narrow this gap by contributing new insights through a comprehensive comparative evaluation study based on a large collection of datasets and models.

3 Research Approach

We perform a series of computational experiments, which we describe in the following. In the first part, we focus on the assessment of the predictive performance, for which we outline the selection of evaluated models (Sect. 3.1), our collection of datasets (Sect. 3.2), and our experimental setup and evaluation pipeline (Sect. 3.3). Thereafter, we consider the models' visual outputs to assess their level of functional interpretability (Sect. 3.4).

3.1 Selection of Models

For our comparative evaluation study, we consider seven different variants of GAMs, for which publicly accessible implementations are available. This includes (i) penalized B-splines (P-Splines) (Eilers and Marx 1996), (ii) penalized thin plate regression splines with shrinkage (TP-Splines) (Wood 2003), (iii) EBM (Nori et al. 2021), (iv) NAM (Agarwal et al. 2021), (v) GAMI-Net (Yang et al. 2021b), (vi) ExNN (Yang et al. 2021a), and (vii) I-GANN (Kraus et al. 2024). The implementations are provided by the respective authors of the proposed GAM variants. For P-Splines, we use the Python package pyGAM (Servén 2021). For TP-Splines, we adopt models from the mgcv package implemented in R (Wood 2023) using a Python wrapper (Chang et al. 2021). Additionally, we also experiment with other spline-based models from the mgcv package, including penalized cubic regression splines with and without shrinkage and different configurations. However, since they only achieve subordinate predictive performance compared to TP-Splines, they are excluded from further model comparison (see Online Appendix E, available online via http://link.springer.com).

Furthermore, we include several baseline models for a broader comparison, which are commonly used in the ML and IS community for prediction tasks using tabular data. Specifically, we include linear models (logistic/linear regression, LR) and decision trees (DTs) as common representatives of intrinsically interpretable models. To consider widely known black-box models, we include RF, XGB, CatBoost, MLP, and TabNet (De Caigny et al. 2024; Schoormann et al. 2023). For the implementation of the models, we use the corresponding Python packages from the scikit-learn library, except for XGB (Chen and Guestrin 2016), CatBoost (Prokhorenkova et al. 2018), and TabNet (Arik and Pfister 2021), where we use Python implementations from the respective developers/authors. Further details on the model implementations can be found in Online Appendix A.

3.2 Selection of Datasets

For an extensive comparison of models, we utilize a variety of benchmark datasets. We align our dataset selection with previous evaluation studies (e.g., Roy et al. 2019; Yang et al. 2021b), using datasets from recognized public repositories such as the UCI Machine Learning Repository[3] and Kaggle[4]. These repositories offer diverse, high-quality datasets for benchmarking purposes, allowing for a thorough evaluation of the different models. After reviewing the repositories, we select twenty diverse datasets that cover a broad range of real-world applications relevant to various business, organizational, and societal issues. To ensure a balanced distribution between different prediction tasks, we use ten datasets tailored for predicting categorical outcomes (binary classification task, CLS) and an equal number for predicting numerical values (regression task, REG). Moreover, we limit our experiments to medium-sized datasets with up to 150,000 samples to maintain a manageable level of computational complexity. This is particularly important given the extensive hyperparameter tuning in our model evaluation procedure (cf. Sect. 3.3).

Table 1 provides an overview of the overall collection with a summary of descriptive statistics of the inherent dataset properties (after cleaning and data preprocessing) and a short description of the prediction targets. The number of samples ranges from 205 to 142,193, and the number of preprocessed and encoded predictors varies between 8 and 100, with a mixed combination of numerical and categorical features. Thus, we consider a variety of benchmark scenarios for assessing and comparing all models. Further details on the data repositories and the applied preprocessing steps can be found in Online Appendix B.

3.3 Experimental Set-Up and Evaluation Pipeline

To provide a fair comparison, we integrate all datasets and models into a shared environment to run the experiments under the same conditions.[5] The predictive performance is measured in terms of commonly used evaluation metrics. For classification tasks, we primarily focus on the area under the receiver operating characteristic curve (AUROC) to assess the models' ability to correctly rank positive and negative instances. For regression tasks, we consider the root mean squared error (RMSE) to quantify the average magnitude of errors between the models' predicted values and the actual values. To determine the best-performing models, we calculate the average ranks across all datasets, with higher ranks (i.e., first, second, third, etc.) indicating better predictive performance. The procedure of ranking algorithms aligns with good practice for evaluating multiple classifiers on multiple datasets (Demšar 2006; Brazdil and Soares 2000).

For the sake of completeness, we also calculate additional metrics that can be found in the online repository, which is referred to at the end of this section.[6] Moreover, we measure

[3] http://archive.ics.uci.edu/ml/ (accessed 20 October 2024).

[4] https://www.kaggle.com/ (accessed 20 October 2024).

[5] In our experiments, we use a workstation with the following setting: Single GPU NVIDIA RTX A 6000 with 48 GB VRAM, Intel i7-12700 10 CPU cores with 20 threads, and 128 GB RAM, Python 3.9.13, Pytorch 1.12.1, and cudatoolkit 11.6.0.

[6] For classification tasks, we additionally measure accuracy, precision, recall, F1-score (macro average, weighted average), and support. For regression tasks, we additionally measure mean squared error (MSE), mean absolute error (MAE), explained variance, r^2, and maximum residual error (max error).

Table 1 Overview of selected datasets covering classification (CLS) and regression (REG) tasks

Type	Dataset	Samples	Features			Prediction target
			num*	cat*	cat**	
CLS	College (Mukti 2022)	1000	4	6	10	Will a high school student go to college?
	Water potability (Kadiwal 2021)	3276	9	0	0	Will the water be safe for consumption?
	Stroke (Palacios 2021)	5110	3	7	16	Will a patient suffer from a stroke?
	Customer churn (IBM 2019)	7043	3	16	37	Will a customer leave the company?
	Recidivism (Larson et al. 2016)	7214	7	4	11	Will a defendant recidivate?
	Credit scoring (Fair Isaac Corporation 2018)	10459	21	2	16	Will a client repay within 2 years?
	Income adults (Kohavi 1996)	32561	6	7	59	Will the income exceed $50,000 per year?
	Bank marketing (Moro et al. 2014)	45211	6	9	41	Will a client subscribe to a deposit?
	Airline satisfaction (Klein 2020)	103904	18	4	6	Will a passenger be satisfied?
	Weather forecast (Bureau of Meteorology 2010)	142193	16	4	54	Will it rain the next day in Australia?
REG	Car price (Kibler et al. 1989)	205	13	11	63	What is the price of a car?
	Student grade (Cortez and Silva 2008)	649	13	17	30	What is a student's final grade?
	Productivity (Imran et al. 2019)	1197	9	4	26	What is the productivity of a team?
	Medical insurance (Lantz 2015)	1338	3	3	6	What are the costs for a patient?
	Violent crimes (Redmond and Baveja 2002)	1994	100	0	0	How many violent crimes will happen?
	Crab farming (Sidhu 2021)	3893	7	1	3	What is the age of a crab?
	Wine quality (Cortez et al. 2009)	4898	11	0	0	What is the quality of a produced wine?
	Bike rental (Fanaee-T and Gama 2014)	17379	7	5	11	How many bikes will be rented per hour?
	House price (Pace and Barry 1997)	20640	8	0	0	What is the value of a house?
	Diamond price (Garside 2021)	53943	6	3	20	What is the price of a diamond?

*After data preprocessing; **after feature encoding

the training time (in seconds) of all models to assess their computational intensity. The results of this examination are reported and discussed in Online Appendix D.

Furthermore, all models are evaluated in two settings. In the first setting, we keep all models in their default hyperparameter configuration. In the second setting, we repeat our evaluation using a grid search tuning our models' hyperparameters out-of-sample. For this purpose, we select the main hyperparameters and establish a grid of parameter values that tend to have a high impact on model performance.

To mitigate the effects of random outliers and to assess the stability and robustness of all models, we apply a well-defined evaluation strategy. We illustrate the strategy in a simplified way in Fig. 2. Specifically, we use a 5-fold (stratified) cross-validation with random shuffling, which allows us to perform error estimation by measuring the out-of-sample performance on each test fold and then calculating the mean and standard deviation across all folds. In addition, for the hyperparameter-tuned setting, we apply an inner-split-validation to tune all hyperparameters. During the random shuffling strategy used in both the cross-validation and the inner-split-validation, seed values are set to ensure reproducible results. Our designed approach is a mild variation of well-established pipelines (De Caigny et al. 2018; Kühl et al. 2021), with modifications justified by the need for comparability and out-of-sample hyperparameter tuning. Overall, this results in 68,500 model runs across all datasets and models, with a total training time exceeding 279 computing hours. Detailed information on the choice of hyperparameters and the best hyperparameter configurations for each dataset and model can be found in Online Appendix C.

To ensure the reproducibility of our experiments, but also to promote the reusability of our results, we incorporate all models, datasets, dataset-agnostic preprocessing steps, metrics, and evaluation procedures into a unified evaluation pipeline, which is publicly available in an online repository.[7] The evaluation pipeline is freely available to researchers and practitioners, and can be easily extended for individual needs (e.g., new models, custom datasets, additional metrics, etc.).

[7] https://github.com/NicoHambauer/Model-Performance-vs-Interpretability (accessed 20 October 2024).

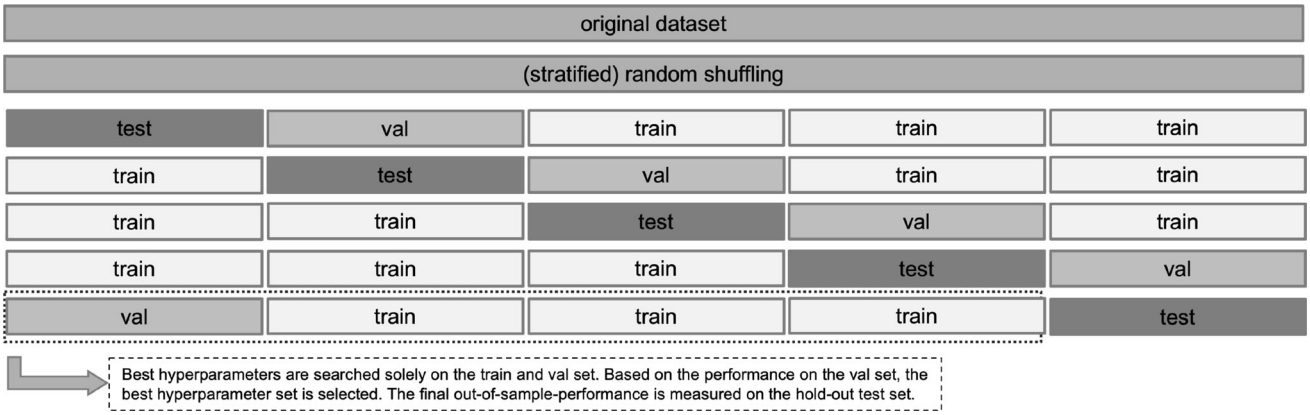

Fig. 2 Illustration of our evaluation strategy using 5-fold (stratified) cross-validation for error estimation and inner train-validation splitting for hyperparameter tuning with final retraining in each iteration

3.4 Assessment of Interpretability

To comprehensively analyze the behavior of all models and assess their level of functional interpretability, our investigations are divided into three parts, offering different perspectives.

In the first part, we consider each model individually and examine the GAMs' general ability to externalize their inner workings through human-understandable feature plots to verify their degree of intrinsic interpretability. More specifically, we follow the notion that an intrinsically interpretable model must be able to provide transparency at the level of individual predictions (i.e., *local interpretability*), but also at the level of the entire model (i.e., *global interpretability*), in order to enable a full understanding of how a model works (Du et al. 2019). To this end, we look into the GAMs' entire output as well as specific shape functions and assess the extent to which each model is able to visually reveal how input features are processed in order to produce a predicted outcome.

In the second part, we contrast and compare the behavior of the different models by looking at them simultaneously. More specifically, we qualitatively examine the GAMs' different feature shape plots across all 20 datasets and models. To ensure a fair and consistent evaluation of the feature plots, we develop custom plot functions for each model and plot type (e.g., for numerical, categorical, and interaction feature effects). On this basis, we identify similarities and differences that allow us to make informed statements about their merits and limitations for different interpretability contexts.

Finally, in the third part, we build on our previous findings and evaluate all models using an objective, aggregated interpretability score. For this purpose, we adopt the six interpretability criteria (see Table 2) suggested by Sudjianto and Zhang (2021), which allow us to reflect on the strengths and weaknesses of each model. Sudjianto and Zhang (2021) propose these six criteria as part of a qualitative assessment framework to determine the level of model-based interpretability in the context of tabular data. For our study, we make some adjustments to the framework to avoid ambiguous evaluations. On the one hand, we adjust the scoring system by simplifying the scales from four levels to three levels. Thus, a model receives two points if it meets a given criterion by default, one point if it meets the criterion after adjusting related hyperparameters, or zero points if a model cannot meet the criterion. On the other hand, we also refine the evaluation criteria in order to increase their conceptual sharpness and to achieve a better common understanding among all authors of this study. For example, the *visualizability* criterion is refined to more clearly emphasize a model's ability to visualize how changes in the model input affect the model output without re-evaluating the model. In Table 2, we summarize the refined evaluation criteria and provide representative examples for each criterion.

The actual evaluation was performed by three of the authors of this paper, who gained extensive experience with each of the models over the course of the study. To ensure an objective assessment, each author independently rated each criterion for each model. The results were then discussed and inter-rater reliability was measured using Krippendorff's alpha (Krippendorff 2018), with an initial agreement of 0.7476. The disagreements were mainly due to different interpretations of the original definitions of the evaluation criteria, which led to the aforementioned refinements. After the refinements, a second round of independent evaluations was performed, resulting in an agreement of 0.9254. The remaining disagreements were due to uncertainties about the effects of certain model hyperparameters to assess whether or not certain interpretability criteria could be met by the corresponding models. To resolve these issues, additional model experiments were performed to reach a final consensus for a correct assessment. Further details on the scoring process,

Table 2 Evaluation criteria for interpretability assessment

Criteria	Descriptions	Examples
Additivity	Whether the model allows for additive decomposition and modular aggregation of individual features so that each feature effect can be analyzed separately	Linear models and GAMs capture features separately and then sum up the feature effects to obtain the overall prediction value
Sparsity	Whether the model uses an internal selection or regularization mechanism to identify a relevant subset of features, with the goal of maximizing their relevance while reducing model complexity	GAMI-Net promotes model compactness by incorporating sparsity constraints to include only non-trivial main effects
Linearity	Whether the model is able to capture linear or piecewise linear relationships between the features and the target, despite the presence of noise in the dataset	IGANN promotes linearity by initializing shape functions linearly and only deviating from this when required by the underlying features
Smoothness	Whether the model avoids erratic curves and abrupt jumps by ensuring that small changes in the input lead to small changes in the output	TP-Splines, GAMI-Net, and IGANN promote smoothness by preserving continuous and gradually changing shape functions
Monotonicity	Whether monotonic constraints can be injected into the model to maintain a steadily increasing or decreasing trend in the feature-target relationships	EBM and GAMI-Net provide functionalities or hyperparameters to impose monotonically increasing or decreasing feature effects
Visualizability	Whether changes in the model's output can be visually understood unambiguously without re-evaluating the overall model when modifying a single feature	P-Splines, TP-Splines, EBM, IGANN, and GAMI-Net produce comprehensible shape plots to visually inspect the impact of individual features

discussion points that arose during the evaluation, and justifications for final scoring decisions can be found in Online Appendix G.

4 Results

4.1 Assessment of Predictive Performance

This section presents the evaluation results for assessing the predictive performance. First, we consider the setting with default model configurations, and then we continue with the results of the models with tuned hyperparameters. Finally, we also consider the performance differences between both settings. The best overall performance for each dataset is highlighted in bold, whereas the best result among the interpretable models is underlined.[8] Moreover, the combination of certain datasets with TP-Splines results in some anomalies where the model does not converge on all five folds, so a subset of folds is used to compute the aggregated performance results. These cases are highlighted in italics.

Default setting Table 3 summarizes the prediction results of the default setting for the classification and regression tasks. The results show that the best prediction scores for each dataset are spread across a variety of models. That is, there is no single model that achieves the best results across all datasets. However, there are some remarkable tendencies. In particular, it can be seen that the intrinsically interpretable GAMs collectively deliver the highest performance in 6 out of 20 datasets. Specifically for classification, this is the case for 4 out of 10 datasets, and for regression, this is the case for 2 out of 10 datasets.

In contrast, the black-box models perform best in 14 out of 20 datasets. Nevertheless, the performance difference between the best black-box models and the best interpretable models is only marginal. For example, for the classification tasks, the smallest difference in performance between the best interpretable model and the best black-box model is 0.003 (college: GAMI-Net 0.969 vs. CatBoost 0.972; income: EBM 0.927 vs. CatBoost 0.930) and the largest difference is 0.012 (weather: EBM 0.875 vs. CatBoost 0.887). Similarly, for the regression tasks, the smallest difference is almost 0 (farming: GAMI-Net 0.655 $\pm$ 0.043 vs. MLP 0.655 $\pm$ 0.042), whereas the largest difference is 0.090 (wine: EBM 0.772 vs. RF 0.682). These results clearly outline that there is no remarkable loss of model accuracy in favor of interpretability.

A more detailed analysis reveals the strengths of the individual models. CatBoost turns out to be the best-performing model overall, showing the best results in 10 out of 20 datasets, with an average rank of 2.93. No other black-box model achieves comparable results. Instead, the strongest competitor is EBM, where it is striking to observe that the interpretable model achieves a remarkable average rank of 4.08. As such, it even dominates the four black-box models MLP, RF, XGB, and TabNet with average ranks of

[8] If multiple models achieve the same average predictive performance values, the model with the lower standard deviation is considered the better one.

Table 3 Predictive performance using default hyperparameters

Task	Dataset	Interpretable models									Black-box models				
		GAMs							Traditional						
		P-Splines	TP-Splines	EBM	NAM	GAMI-Net	ExNN	IGANN	LR	DT	RF	XGB	CatBoost	MLP	TabNet
CLS	college	0.951	0.951	0.965	0.889	0.969	0.752	0.952	0.935	0.839	0.964	0.962	**0.972**	0.968	0.949
		(0.004)	(0.006)	(0.008)	(0.025)	(0.008)	(0.033)	(0.008)	(0.012)	(0.025)	(0.011)	(0.015)	(0.007)	(0.013)	(0.013)
	water	0.561	0.585	0.678	0.564	0.679	0.680	0.573	0.500	0.562	0.686	0.656	**0.689**	0.681	0.636
		(0.016)	(0.015)	(0.026)	(0.021)	(0.034)	(0.025)	(0.033)	(0.016)	(0.016)	(0.017)	(0.014)	(0.017)	(0.031)	(0.020)
	stroke	0.834	0.835	0.837	0.802	0.833	0.702	**0.838**	0.837	0.560	0.800	0.803	0.806	0.610	0.805
		(0.022)	(0.024)	(0.017)	(0.020)	(0.023)	(0.034)	(0.027)	(0.026)	(0.012)	(0.018)	(0.011)	(0.015)	(0.197)	(0.051)
	churn	**0.850**	0.849	0.849	0.827	0.847	0.774	0.847	0.845	0.662	0.821	0.824	0.837	0.842	0.835
		(0.011)	(0.012)	(0.011)	(0.008)	(0.014)	(0.020)	(0.014)	(0.015)	(0.015)	(0.013)	(0.010)	(0.012)	(0.016)	(0.015)
	recidivism	0.744	0.744	**0.748**	0.705	0.741	0.634	0.739	0.727	0.602	0.674	0.724	0.740	0.735	0.727
		(0.017)	(0.018)	(0.016)	(0.026)	(0.018)	(0.014)	(0.019)	(0.017)	(0.012)	(0.015)	(0.017)	(0.015)	(0.012)	(0.016)
	credit	0.806	***0.810***	0.808	0.791	0.804	0.783	0.802	0.797	0.636	0.797	0.781	0.794	0.799	0.785
		(0.008)	(0.010)	(0.010)	(0.012)	(0.010)	(0.017)	(0.010)	(0.012)	(0.015)	(0.011)	(0.008)	(0.007)	(0.010)	(0.013)
	income	0.916	*0.916*	0.927	0.897	0.914	0.609	0.915	0.907	0.748	0.903	0.927	**0.930**	0.913	0.908
		(0.002)	*(0.002)*	(0.002)	(0.002)	(0.002)	(0.008)	(0.002)	(0.002)	(0.002)	(0.004)	(0.001)	(0.002)	(0.003)	(0.004)
	bank	0.780	0.779	0.797	0.758	0.780	0.615	0.774	0.765	0.613	0.781	0.790	**0.805**	0.782	0.775
		(0.011)	(0.011)	(0.011)	(0.013)	(0.011)	(0.009)	(0.012)	(0.009)	(0.006)	(0.011)	0.009)	(0.009)	(0.011)	(0.007)
	airline	0.980	0.980	0.989	0.963	0.985	0.990	0.980	0.927	0.944	0.993	**0.995**	**0.995**	0.994	0.993
		(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.002)	(0.001)	(0.002)	(0.001)	(0.000)	(0.000)	(0.000)	(0.001)	(0.002)
	weather	0.874	0.874	0.875	0.852	0.875	0.579	0.873	0.868	0.696	0.883	0.886	**0.887**	0.885	0.878
		(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.005)	(0.002)	(0.003)	(0.002)	(0.003)	(0.003)
REG	car	0.367	*0.405*	**0.256**	0.715	0.381	1.712	0.350	0.470	0.357	0.304	0.301	0.288	0.317	0.437
		(0.105)	*(0.113)*	(0.071)	(0.399)	(0.139)	(0.899)	(0.179)	(0.263)	(0.102)	(0.095)	(0.071)	(0.091)	(0.076)	(0.236)
	student	0.882	0.863	0.856	0.954	0.900	1.682	0.850	0.878	1.238	0.842	0.883	**0.832**	0.876	0.921
		(0.121)	(0.132)	(0.127)	(0.154)	(0.170)	(0.386)	(0.116)	(0.151)	(0.135)	(0.143)	(0.120)	(0.146)	(0.146)	(0.093)
	productivity	0.796	0.789	0.723	0.909	0.742	1.036	0.813	0.905	0.960	0.707	0.744	**0.698**	0.784	0.902
		(0.093)	(0.081)	(0.048)	(0.054)	(0.052)	(0.060)	(0.076)	(0.043)	(0.085)	(0.065)	(0.042)	(0.036)	(0.055)	(0.024)
	insurance	0.500	0.497	**0.379**	0.683	0.383	0.625	0.495	0.863	0.523	0.400	0.435	0.388	0.397	0.405
		(0.020)	(0.019)	(0.027)	(0.028)	(0.026)	(0.020)	(0.020)	(0.071)	(0.036)	(0.021)	(0.019)	(0.027)	(0.016)	(0.031)
	crimes	0.737	0.586	0.590	0.774	0.610	0.684	0.584	0.600	0.844	0.595	0.623	**0.571**	0.612	0.642
		(0.047)	(0.059)	(0.065)	(0.048)	(0.059)	(0.050)	(0.054)	(0.066)	(0.072)	(0.061)	(0.048)	(0.061)	(0.062)	(0.067)
	farming	0.673	0.665	0.687	0.793	0.655	0.668	0.668	0.798	0.931	0.682	0.725	0.673	**0.655**	0.677
		(0.042)	(0.043)	(0.043)	(0.036)	(0.043)	(0.026)	(0.046)	(0.048)	(0.041)	(0.033)	(0.040)	(0.037)	(0.042)	(0.052)
	wine	0.828	0.815	0.772	0.880	0.798	0.811	0.813	0.893	0.966	**0.682**	0.707	0.729	0.777	0.803
		(0.046)	(0.026)	(0.021)	(0.037)	(0.019)	(0.019)	(0.024)	(0.028)	(0.037)	(0.032)	(0.027)	(0.022)	(0.019)	(0.020)

Table 3 continued

Task	Dataset	Interpretable models									Black-box models				
		GAMs							Traditional						
		P-Splines	TP-Splines	EBM	NAM	GAMI-Net	ExNN	IGANN	LR	DT	RF	XGB	CatBoost	MLP	TabNet
	bike	0.553	0.581	0.315	0.748	0.306	0.913	0.636	0.830	0.325	0.235	**0.227**	0.230	0.255	0.311
		(0.010)	(0.013)	(0.006)	(0.076)	(0.009)	(0.018)	(0.016)	(0.020)	(0.008)	(0.008)	(0.006)	(0.007)	(0.006)	(0.022)
	house	0.524	0.536	0.445	0.704	0.477	0.501	0.547	0.771	0.599	0.424	0.415	**0.411**	0.465	0.471
		(0.017)	(0.018)	(0.014)	(0.029)	(0.015)	(0.015)	(0.018)	(0.009)	(0.018)	(0.014)	(0.011)	(0.014)	(0.013)	(0.016)
	diamond	0.383	0.634	0.230	0.413	0.214	0.980	0.268	0.471	0.189	0.138	**0.138**	0.139	0.147	0.148
		(0.154)	(0.724)	(0.010)	(0.026)	(0.009)	(0.013)	(0.008)	(0.011)	(0.008)	(0.004)	(0.003)	(0.004)	(0.018)	(0.005)
Average rank CLS		6.15	6.00	3.55	11.20	5.35	11.50	6.95	9.70	13.40	7.30	6.45	3.55	5.70	8.20
Average rank REG		9.20	8.00	4.60	12.30	5.90	11.30	7.30	11.50	11.20	3.60	5.00	2.30	4.80	8.00
Average rank total		7.68	7.00	4.08	11.75	5.63	11.40	7.13	10.60	12.30	5.45	5.73	2.93	5.25	8.10

Classification tasks are assessed using AUROC, whereas regression tasks are measured using RMSE. The first value in each cell represents the mean value over five folds. The second value represents the standard deviation. The best overall performance for each dataset is highlighted in bold, whereas the best result among the interpretable models is underlined

5.25, 5.45, 5.73, and 8.10, respectively. If we look only at the results of the classification tasks, EBM takes the top position together with CatBoost, where both models achieve an average rank of 3.55.

The second best interpretable model is GAMI-Net with an average rank of 5.58, followed by TP-Splines, IGANN, and P-Splines with average ranks of 7.00, 7.13, and 7.68, respectively. Thus, all four models also show strong prediction qualities that are not too far behind their black-box counterparts. The remaining interpretable models NAM, ExNN, LR, and DT generally underperform in this setting, as indicated by their inferior average ranks.

In summary, the results challenge the prevalent tendency in the ML community to favor black-box models over interpretable alternatives, which in fact clearly show competitive performance, as our evaluation results reveal.

Tuned setting Next, we focus on the prediction results of the hyperparameter-tuned models, obtained via a grid search.[9] Table 4 summarizes the results of the evaluation. Again, we can see that several intrinsically interpretable models, such as EBM, IGANN, and GAMI-Net, are characterized by strong prediction qualities. Similar to the default setting, the interpretable models perform best in 6 out of 20 datasets, whereas the black-box models show better results in 14 out of 20 datasets.

Although the black-box models achieve some performance gains in the tuned setting, the difference in performance between the best models from both groups remains comparatively small. For the classification tasks, the smallest difference in performance is 0.001 (income: EBM 0.928 vs. XGB/CatBoost 0.929) and the largest difference is 0.015 (weather: EBM 0.878 vs. CatBoost 0.893). For the regression tasks, the smallest difference is 0.003 (student: EBM 0.843 vs. RF 0.840), whereas the largest difference is 0.097 (bike: EBM 0.296 vs. CatBoost 0.199).

Looking at the individual models, it is remarkable that EBM still remains the second best-performing model overall with an average rank of 4.05, just behind CatBoost with an average rank of 2.85. For the classification datasets, the interpretable model even slightly outperforms CatBoost by achieving the best average rank across all models with a score of 3.35. Furthermore, EBM shows the highest predictive performance across all models in 4 out of 20 datasets and outperforms its interpretable competitors in 14 out of 20 datasets. The next best models after CatBoost and EBM are XGB and RF with average ranks of 4.30 and 4.83, respectively. This is followed by IGANN as another interpretable model, with an average rank of 6.30 outperforming even the two black-box models MLP and TabNet with average ranks of 6.48 and 9.75, respectively.

[9] The corresponding hyperparameter configurations for each dataset and model can be found in Online Appendix C.

Table 4 Predictive performance using tuned hyperparameters

Task	Dataset	Interpretable Models									Black-box Models				
		GAMs							Traditional						
		P-Splines	TP-Splines	EBM	NAM	GAMI-Net	ExNN	IGANN	LR	DT	RF	XGB	CatBoost	MLP	TabNet
CLS	college	0.951	0.952	0.967	0.935	0.967	0.768	0.953	0.929	0.849	0.964	0.960	0.970	**0.973**	0.944
		(0.006)	(0.007)	(0.008)	(0.011)	(0.008)	(0.025)	(0.023)	(0.015)	(0.027)	(0.012)	(0.017)	(0.008)	(0.009)	(0.012)
	water	0.565	0.589	0.679	0.578	0.675	0.680	0.664	0.495	0.589	0.684	0.658	**0.688**	0.677	0.629
		(0.019)	(0.018)	(0.027)	(0.023)	(0.030)	(0.017)	(0.049)	(0.013)	(0.027)	(0.015)	(0.011)	(0.021)	(0.044)	(0.040)
	stroke	0.835	*0.828*	0.834	0.804	0.834	0.701	**0.838**	0.834	0.815	0.836	0.823	0.830	0.662	0.795
		(0.023)	*(0.022)*	(0.017)	(0.044)	(0.026)	(0.018)	(0.027)	(0.032)	(0.035)	(0.018)	(0.027)	(0.019)	(0.176)	(0.037)
	churn	0.849	0.849	**0.849**	0.846	0.847	0.776	0.846	0.838	0.823	0.841	0.844	0.848	0.842	0.832
		(0.014)	(0.013)	(0.011)	(0.014)	(0.015)	(0.017)	(0.013)	(0.014)	(0.016)	(0.013)	(0.014)	(0.012)	(0.018)	(0.016)
	recidivism	0.741	0.743	**0.747**	0.742	0.740	0.632	0.744	0.726	0.709	0.736	0.744	0.742	0.740	0.722
		(0.019)	(0.017)	(0.017)	(0.017)	(0.017)	(0.014)	(0.019)	(0.017)	(0.015)	(0.016)	(0.018)	(0.026)	(0.017)	(0.016)
	credit	0.807	**0.810**	0.807	0.802	0.804	0.787	0.802	0.795	0.768	0.801	0.802	0.797	0.796	0.779
		(0.009)	(0.008)	(0.010)	(0.009)	(0.010)	(0.017)	(0.010)	(0.013)	(0.011)	(0.011)	(0.012)	(0.006)	(0.009)	(0.021)
	income	0.916	*0.920*	0.928	0.916	0.915	0.609	0.916	0.906	0.902	0.917	**0.929**	**0.929**	0.915	0.906
		(0.002)	*(0.003)*	(0.002)	(0.001)	(0.002)	(0.007)	(0.002)	(0.002)	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)
	bank	0.777	0.778	0.798	0.772	0.772	0.609	0.772	0.764	0.719	0.795	0.797	**0.801**	0.787	0.747
		(0.011)	(0.009)	(0.011)	(0.014)	(0.014)	(0.008)	(0.007)	(0.008)	(0.026)	(0.006)	(0.008)	(0.006)	(0.009)	(0.016)
	airline	0.980	0.980	0.991	0.976	0.989	0.991	0.989	0.927	0.958	0.994	**0.995**	**0.995**	0.994	0.994
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.003)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	weather	0.874	0.874	0.878	0.870	0.876	0.573	0.878	0.868	0.840	0.887	0.891	**0.893**	0.887	0.877
		(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.005)	(0.003)	(0.003)	(0.001)	(0.002)	(0.003)	(0.002)	(0.002)	(0.002)
REG	car	0.293	0.343	**0.262**	0.422	0.366	1.689	0.345	0.326	0.346	0.309	0.291	0.286	0.330	0.687
		(0.124)	(0.064)	(0.084)	(0.241)	(0.145)	(0.613)	(0.166)	(0.097)	(0.102)	(0.093)	(0.077)	(0.086)	(0.087)	(0.458)
	student	0.862	0.856	0.843	0.850	0.873	1.675	0.861	0.868	0.885	**0.840**	0.853	0.841	0.915	0.927
		(0.136)	(0.132)	(0.134)	(0.134)	(0.113)	(0.251)	(0.123)	(0.147)	(0.136)	(0.140)	(0.142)	(0.153)	(0.147)	(0.169)
	productivity	0.832	0.768	0.716	0.859	0.746	1.023	0.775	0.846	0.842	0.705	0.693	**0.689**	0.773	0.848
		(0.096)	(0.060)	(0.037)	(0.030)	(0.053)	(0.049)	(0.050)	(0.031)	(0.046)	(0.054)	(0.034)	(0.033)	(0.041)	(0.055)
	insurance	0.501	0.499	0.377	0.509	0.381	0.618	0.381	0.503	0.405	0.378	**0.373**	0.373	0.397	0.443
		(0.019)	(0.018)	(0.026)	(0.023)	(0.023)	(0.021)	(0.021)	(0.018)	(0.016)	(0.025)	(0.027)	(0.033)	(0.020)	(0.049)
	crimes	0.575	**0.572**	0.585	0.607	0.585	0.597	0.581	0.587	0.676	0.595	0.592	0.577	0.596	0.629
		(0.056)	(0.061)	(0.065)	(0.052)	(0.055)	(0.081)	(0.056)	(0.062)	(0.047)	(0.062)	(0.064)	(0.058)	(0.060)	(0.061)
	farming	0.665	0.667	0.685	0.708	0.657	0.666	0.656	0.693	0.729	0.674	0.678	0.677	**0.650**	0.666
		(0.043)	(0.042)	(0.044)	(0.041)	(0.043)	(0.031)	(0.044)	(0.045)	(0.059)	(0.038)	(0.039)	(0.045)	(0.042)	(0.050)
	wine	0.828	0.898	0.765	0.822	0.801	0.813	0.795	0.852	0.845	0.678	0.686	**0.669**	0.779	0.814
		(0.038)	(0.194)	(0.021)	(0.019)	(0.022)	(0.031)	(0.026)	(0.027)	(0.029)	(0.032)	(0.031)	(0.028)	(0.021)	(0.026)
	bike	0.551	0.553	0.296	0.570	0.297	0.914	0.408	0.776	0.325	0.234	0.214	**0.199**	0.224	0.320
		(0.009)	(0.010)	(0.006)	(0.014)	(0.007)	(0.017)	(0.009)	(0.017)	(0.006)	(0.008)	(0.006)	(0.006)	(0.009)	(0.041)

Table 4 continued

Task	Dataset	Interpretable Models									Black-box Models				
		GAMs							Traditional						
		P-Splines	TP-Splines	EBM	NAM	GAMI-Net	ExNN	IGANN	LR	DT	RF	XGB	CatBoost	MLP	TabNet
	housing	0.521	0.522	0.439	0.555	0.478	0.501	0.490	0.603	0.539	0.422	0.397	**0.387**	0.442	0.461
		(0.016)	(0.016)	(0.012)	(0.013)	(0.016)	(0.024)	(0.015)	(0.020)	(0.017)	(0.014)	(0.011)	(0.010)	(0.009)	(0.012)
	diamond	0.433	2.122	0.207	0.268	0.217	0.980	0.212	0.284	0.182	0.137	**0.134**	0.134	0.136	0.168
		(0.289)	(3.271)	(0.012)	(0.009)	(0.005)	(0.013)	(0.007)	(0.007)	(0.004)	(0.003)	(0.002)	(0.004)	(0.003)	(0.031)
Average rank CLS		7.15	6.30	3.35	8.95	6.75	11.85	5.95	11.35	12.60	5.35	4.80	3.40	6.95	10.25
Average rank REG		8.10	8.80	4.75	11.20	6.90	11.65	6.65	10.80	10.50	4.30	3.80	2.30	6.00	9.25
Average rank total		7.63	7.55	4.05	10.08	6.83	11.75	6.30	11.08	11.55	4.83	4.30	2.85	6.48	9.75

Classification tasks are assessed using AUROC, whereas regression tasks are measured using RMSE. The first value in each cell represents the mean value over five folds. The second value represents the standard deviation. The best overall performance for each dataset is highlighted in bold, whereas the best result among the interpretable models is underlined

GAMI-Net, TP-Splines, and P-Splines rank in the middle with average ranks of 6.83, 7.55, and 7.63, respectively, while the remaining models NAM, ExNN, LR, and DT again exhibit inferior performance.

Moreover, if we compare the best interpretable model EBM with the best black-box models CatBoost, XGB, and RF, the difference in predictive performance is mostly in the range of < 0.01. Only for some larger regression datasets, the three black-box models have slight advantages with differences in the range of < 0.099. Overall, however, the results show that the performance difference between the interpretable GAMs and the black-box models is marginally small.

Predictive performance difference between default and tuned models To shift the perspective towards a better understanding of the change in predictive performance resulting from the hyperparameter tuning, we summarize the performance difference between the default setting and the tuned setting in Table 5. For better readability, we only report values that are greater than the standard deviation of the tuned models.

The results reveal several insightful observations. For example, the majority of GAMs, including EBM, P-Splines, TP-Splines, GAMI-Net, and ExNN, generally show little or no improvement. This illustrates that the models in their default configurations already produce fairly robust results without the need for extensive hyperparameter tuning. In the case of IGANN, we can see some improvements, especially for larger datasets. In contrast, the largest differences can be noted for NAM. Here, we find improvements in almost all datasets, indicating that NAM is very tuning-intensive and must be adapted to the particular circumstances of the datasets.

Similarly, several of the traditional models, including LR, DT, RF, and XGB, require additional hyperparameter tuning to unfold better prediction qualities. While LR shows notable improvements only for regression tasks, the situation is reversed for RF, which shows improvements only for classification tasks. In contrast, DT improves the most among all four models, showing performance gains on almost all datasets. This is not surprising, as decision trees generally tend to overfit without explicit model tuning. Similarly, XGB shows improvements on almost all datasets. As a result, the average ranks of RF and XGB move much closer to the average ranks of the best-performing models CatBoost and EBM. Nevertheless, it is striking to observe that especially the intrinsically interpretable model EBM still dominates the majority of black-box models even without the need of additional performance gains through extensive hyperparameter tuning.

In summary, we find that the results remain relatively consistent in both settings. Looking at the rank differences at the bottom of Table 5, we see a slight improvement of

Table 5 Difference in models' predictive performance between tuned and default hyperparameters

Task	Dataset	Interpretable models									Black-box models				
		GAMs							Traditional						
		P-Splines	TP-Splines	EBM	NAM	GAMI-Net	ExNN	IGANN	LR	DT	RF	XGB	CatBoost	MLP	TabNet
CLS	college				0.046										
	water							0.091							
	stroke									0.255	0.036		0.024		
	churn				0.019					0.161	0.020	0.020			
	recidivism				0.037					0.107	0.062	0.020			
	credit				0.011					0.132		0.021			
	income		*0.004*		0.019					0.154	0.014	0.002		0.002	− 0.002
	bank				0.014					0.106	0.014				− 0.028
	airline			0.002	0.013	0.004	0.001	0.009		0.014	0.001				0.001
	weather			0.003	0.018		− 0.006	0.005		0.144	0.004	0.005	0.006	0.002	
REG	car				− 0.293				− 0.144						
	student									− 0.353					
	productivity				− 0.050				− 0.059	− 0.118		− 0.051			
	insurance				− 0.174			− 0.114	− 0.360	− 0.118		− 0.062			
	crimes	− 0.162			− 0.167		− 0.087			− 0.168					
	farming				− 0.085				− 0.105	− 0.202		− 0.047			
	wine				− 0.058				− 0.041	− 0.121			− 0.060		
	bike		− 0.028	− 0.019	− 0.178	− 0.009		− 0.228	− 0.054			− 0.013	− 0.031	− 0.031	
	housing				− 0.149			− 0.057	− 0.168	− 0.060		− 0.018	− 0.024	− 0.023	
	diamond			−0.023	−0.145			−0.056	−0.187	−0.007		−0.004	−0.005	−0.011	
Default average rank		7.68	7.00	4.08	11.75	5.63	11.40	7.13	10.60	12.30	5.45	5.73	2.93	5.25	8.10
Tuned average rank		7.63	7.55	4.05	10.08	6.83	11.75	6.30	11.08	11.55	4.83	4.30	2.85	6.48	9.75
Rank difference		↑ 0.05	↓ 0.55	↑ 0.03	↑ 1.68	↓ 1.20	↓ 0.35	↑ 0.83	↓ 0.48	↑ 0.75	↑ 0.63	↑ 1.43	↑ 0.07	↓ 1.23	↓ 1.65

Only changes that deviate beyond the standard deviation are reported. Note that a positive value for classification tasks represents a performance improvement as the difference is measured via AUROC, whereas for regression tasks a negative value indicates an improvement as the difference is measured via RMSE

about 1.5 average ranks for XGB and NAM, while GAMI-Net, MLP, and TabNet decline by approximately 1.5 average ranks. Despite these variations, the overall performance ratios remain relatively stable. Therefore, we do not observe a remarkable performance advantage in favor of the black-box models, which is often a widespread belief according to the assumption of the performance-interpretability trade-off (Rudin 2019). In fact, the opposite seems to be true, as the interpretable models demonstrate competitive performance levels while offering transparent model structures. We delve deeper into this aspect in the next section.

4.2 Assessment of Model Interpretability

As described in Sect. 3.4, the interpretability assessment is divided into three parts. First, we consider the GAMs' ability to externalize their inner workings by offering locally and globally interpretable results. Then, we compare the visual output of the different models to highlight similarities and differences between them. Finally, we evaluate the models' level of interpretability using the objective evaluation criteria introduced above in Table 2.

Global and local interpretability For the first assessment, we consider an EBM model and a reduced version of the *bike* dataset to illustrate the expressiveness of the feature plots in terms of their global and local interpretability. Although we specifically use EBM for our demonstration (because it provides the best results among all interpretable models for this particular dataset), it is important to emphasize that it only serves as one representative example of the entire GAM family. The demonstration provides insight into the inherent properties of GAMs as a whole, and is broadly applicable to the other models as well. An exception, however, are the ExNN and NAM models, which show a different behavior and therefore are discussed separately.

The *bike* dataset contains about 17,000 samples of hourly counts of rented bikes from a bike-sharing system, along with 12 features that include details about weather and seasonal information (Fanaee-T and Gama 2014). As such, we consider a regression task with the goal of predicting the number of bikes rented per hour. For our demonstration, we trained a model with a filtered feature set to provide a holistic discussion of the different global and local interpretation perspectives. Specifically, we consider two numerical features (perceived temperature: *atemp*, hour of the day: *hr*), one categorical feature (weather situation: *weathersit*), and two pairwise interaction terms that are automatically identified by the EBM model (*atemp* × *hr*, *weathersit_3* × *hr*).

After fitting the model for the regression task, it offers several feature plots that visualize the captured relationships between the input features and the predicted number of rented bikes, as summarized in Fig. 3. The first plot displays the model's *feature importance*, indicating that *hr* is the most important feature and *weathersit_4* is the least important feature. The remaining plots show the learned *shape functions* of the model, revealing how each value of the input features affects the predicted output.

Depending on the feature types, different shape plots are generated. The effects of numerical features are shown by line plots with curved shapes (cf. *hr* and *atemp*), whereas the effects of categorical features are shown by bar plots (cf. *weathersit*). The x-axes represent the feature values and the y-axes represent their impact on the target variable compared to the average prediction value (i.e., positive impact for $y > 0$, negative impact for $y < 0$). For example, it can be seen that the most important feature *hr* has a strong positive impact with x-values around 9 and 17, whereas feature values below 7 generally have a negative impact on the predicted output. In other words, the model captures the likely relationship that many bikes will be rented during the morning and afternoon rush hours, whereas at night the number strongly declines.

The shape plots for pairwise interactions are slightly different. For interaction terms between a numerical and categorical feature, a line plot is used with different lines representing the distinct groups of the categorical feature (cf. *weathersit_3* × *hr*). For interaction terms between two numerical features, both the x-axis and the y-axis represent the feature values, whereas the impact on the target is highlighted with a corresponding color scheme, resulting in a two-dimensional heatmap (cf. *atemp* × *hr*). For example, it can be seen that low temperatures during the early evening hours have a strong negative impact on the predicted outcome (i.e., dark blue area), whereas medium to high temperatures during the same time period have a strong positive impact on the predicted outcome (i.e., light yellow area).

Taken together, the shape plots collectively externalize the model's inner decision logic and visualize how each feature affects the prediction target over its entire range of values. Thus, they address the demand of providing *intrinsic global interpretability*, which is necessary to understand the model's behavior as a whole and to identify potential biases and other pitfalls (Du et al. 2019; Rudin 2019).

Apart from that, it is also possible to replicate and verify the prediction scores for individual samples by extracting instance-based computations. This procedure is indicated by the red dots in Fig. 3. The sum based on the x and y values is displayed in the gray box in the lower part of Fig. 3, highlighting the model's functionality to provide *intrinsic local interpretability*. This is useful to understand the reasoning behind individual predictions, or why a

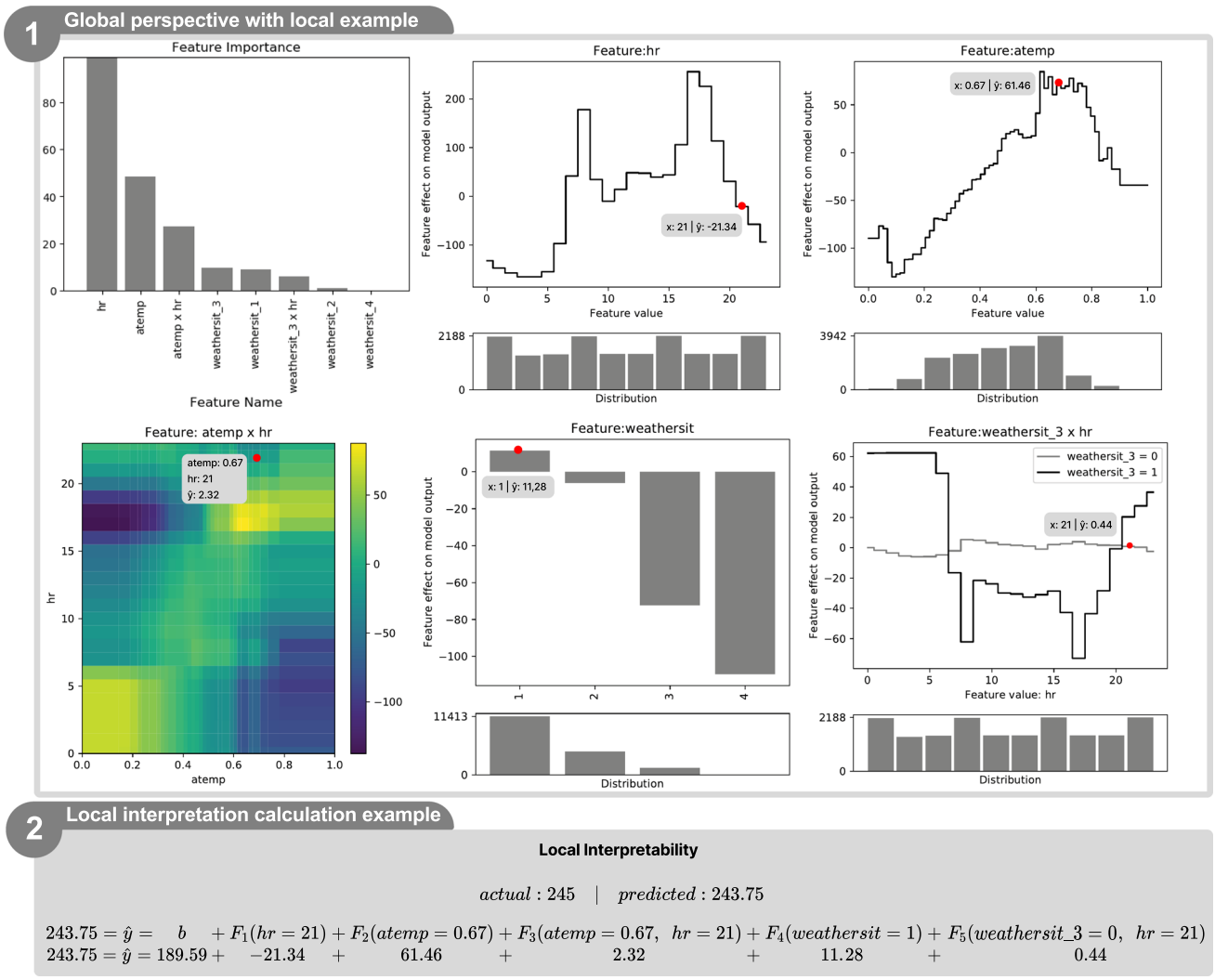

Fig. 3 Global and local interpretability of an EBM model. (1) The summary plots visualize the overall feature importance and the global relationships between the input features and the prediction target. The red dots illustrate the possibility of reading local values of individual feature effects for an exemplary sample. (2) The corresponding calculation example shows the additive contributions of each feature to the model output to retrieve the final prediction for the selected sample

particular prediction was made for a given input. Furthermore, the example shows that the shape plots are not just an approximate explanation of relevant relationships, but an *exact* description of how the intrinsically interpretable model computes a prediction. This is a unique property of the GAMs, which, unlike complex black-box models, add up each feature effect to obtain a transparently derived prediction value for comprehensible decision support.

Comparison of shape plots between GAMs After verifying the models' global and local interpretability, we continue to examine similarities and variations between the visual outputs of the different GAMs, which are mostly reflected in the appearance of their shape plots. On this basis, we can derive statements regarding their merits and limitations for different interpretability contexts.

Figure 4 summarizes exemplary results for five GAMs that are roughly similar but differ in detail, including IGANN, GAMI-Net, TP-Splines, P-Splines, and EBM.[10] The results of the different models are shown in the rows of the grid. Additionally, we have included the output of the linear model (i.e., LR), which serves as a reference line. The columns of the grid in Fig. 4 represent selected features from four different datasets, with a brief description of the features and the prediction targets.

[10] Note that most GAMs are capable of producing confidence bands to provide insight into model uncertainty/variability in certain feature regions. However, this functionality was neglected in Fig. 4 to focus on the visual properties of the different shape functions. For completeness, another version of the figure with confidence bands is provided in Online Appendix F.

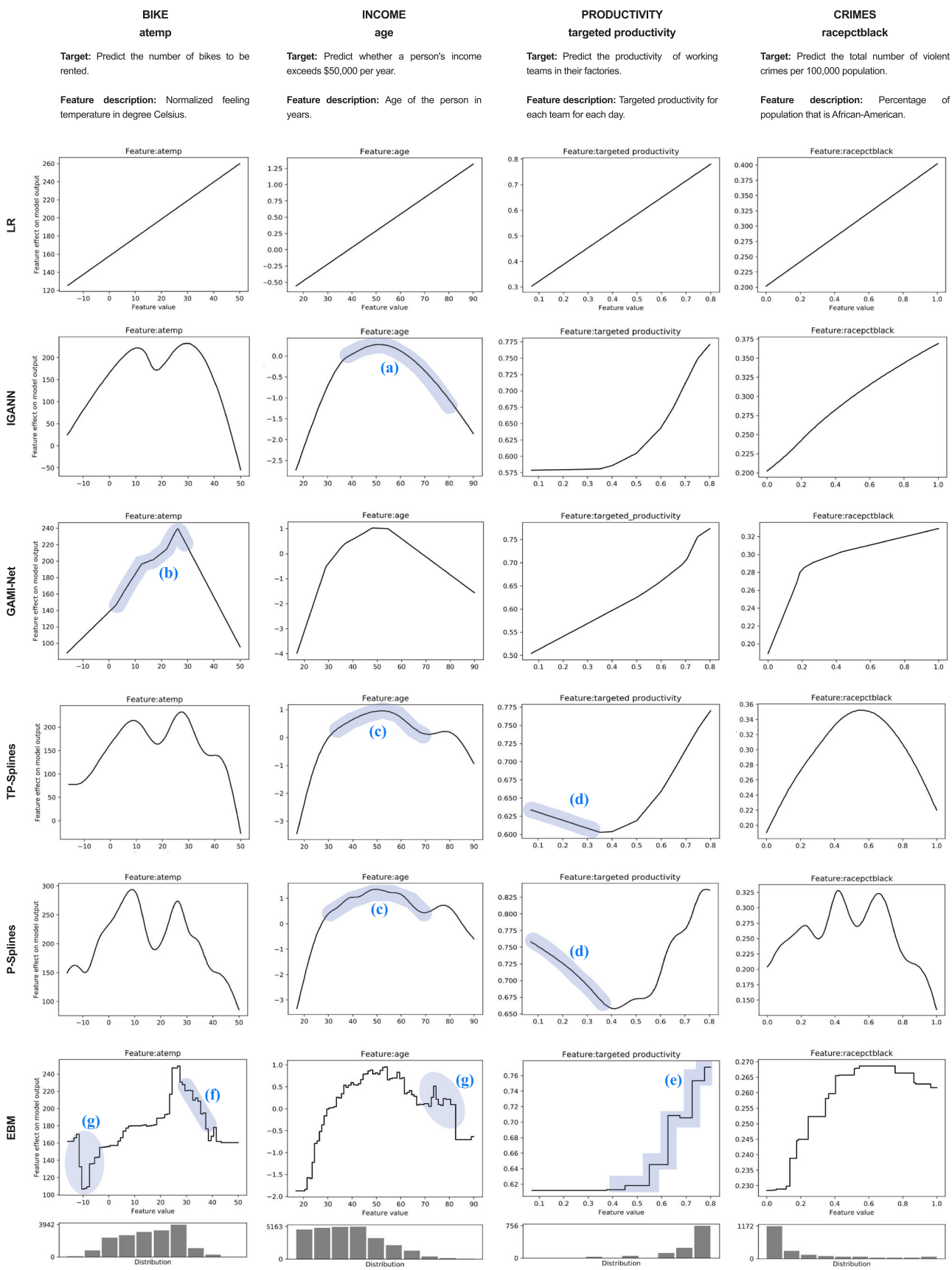

Fig. 4 Comparison of shape plots learned by five different GAMs and a linear model

As expected, the linear model yields the simplest relationships that are easy to comprehend. At the same time, however, the linearity constraint impairs the model's ability to capture more complex feature characteristics, resulting in inferior predictive performance. In contrast, all four GAMs generally agree on non-linear relationships, with roughly comparable feature representations. However, their different model structures also lead to observable differences.

For example, IGANN produces fairly smooth shape functions that promote interpretability by avoiding sharp and erratic step functions and minor details, as is the case with other models (e.g., see the light blue area (a) in the *income* dataset for IGANN compared to the same area of all other GAMs). This behavior is due to IGANN's principle of initializing the training procedure with linear functions and then gradually adapting to non-linearities if the underlying data requires it (Kraus et al. 2024). Interestingly, despite neglecting minor details in the shape functions, IGANN achieves very good performance scores with a ranking at least among the top five tuned models, as the results of the previous section indicate. Thus, it can be concluded that IGANN is less prone to overfitting as it does not capture too much noise.

A similar behavior can also be observed with GAMI-Net, which captures smooth shape functions that avoid sharp jumps and small details while positioning among the best-performing (interpretable) models. Besides that, GAMI-Net is able to produce piecewise linear shapes using a ReLU activation function, as highlighted in area (b) of the *bike* dataset. This ensures a high degree of interpretability as it is easy to comprehend how changes in the feature values affect the target variable in different feature regions. Another strength of GAMI-Net is that it focuses on sparsity and regularization constraints (Yang et al. 2021b), which makes it particularly interesting for high-dimensional datasets with many features involved.

In contrast, the shape functions learned by the spline-based GAMs are generally characterized by more detail and a lower degree of smoothness. For example, while TP-Splines use additional penalization and smoothing techniques, P-Splines are just a combination of piecewise polynomial basis functions (i.e., B-splines) with a discrete penalty on the basis coefficients (Eilers and Marx 1996). Therefore, TP-Splines generally result in smoother shape functions compared to P-Splines (cf. highlighted areas (c)). Nevertheless, the degree of smoothness can also be controlled by several hyperparameters, such as the number of splines. That is, more splines generally mean curvier shape functions (see Online Appendix F). Moreover, it can be observed that both P-Splines and TP-Splines tend to produce overly confident extrapolations in feature regions with small sample sizes (Chang et al. 2021). Some examples of such extrapolations can be seen in the highlighted areas (d) of the *productivity* dataset, where both models capture the misleading relationship that a lower level of targeted productivity increases the likelihood of a higher level of actual productivity, which may lead to erroneous conclusions.

The EBM is located at the other end of the spectrum. It generally produces detailed shape functions that are most similar to those of the spline-based models. However, because EBM is based on tree ensembles, the shapes are characterized by piecewise constant step functions (Lou et al. 2012). These functions undergo abrupt changes and sharp jumps at discrete values, which is highly favorable in situations where features inherently exhibit step-like patterns. Such an effect can be seen in the highlighted area (e), which shows the feature effects for different discrete productivity levels. In other situations, however, such sharp jumps are hard to justify for decision-making purposes (cf. highlighted area (f) that shows an erratic downward trend for the feature effect). Similarly, the model tends to capture even subtle nuances in the data (cf. highlighted areas (g)), which further impairs the model's interpretability. Nevertheless, such detailed patterns can also have positive implications, such as detecting anomalies that aid in removing data quality issues or support model debugging purposes. Furthermore, this comprehensive approach appears to be effective, as the EBM model generally shows the strongest predictive performance in comparison to CatBoost, XGB, and other opaque models.

In contrast to the five GAMs discussed above, we excluded NAM and ExNN from our previous considerations. This is because they show a fundamentally different behavior. NAM suffers from strong overfitting in all our experiments, which could not be resolved even with extensive hyperparameter tuning and additional model implementations. As a result, the model exhibits below-average performance and produces extremely jagged shape functions that are hardly able to convey a comprehensible decision logic. Further details as well as exemplary shape plots for the *bike* and the *productivity* datasets can be found in Online Appendix F.

The ExNN, on the other hand, is a special case, as it is based on the structure of an additive index model. For the interpretation of the model, this means that not only a single feature is covered by a corresponding shape function, but that a whole set of features may provide partial contributions to that shape function. Consequently, the model output is barely interpretable in a meaningful way when many features are

[11] It could be argued that LR is not sparse by default, since regularization constraints must be explicitly considered. However, in our experiments, we use LogisticRegression and ElasticNet from the sklearn (Python) package, which explicitly consider L1/L2 regularization.

Table 6 Results of the interpretability assessment

Task	Interpretable models									Black-box models				
	GAMs							Traditional						
	P-Splines	TP-Splines	EBM	NAM	GAMI-Net	ExNN	IGANN	LR	DT	RF	XGB	CatBoost	MLP	TabNet
Additivity	2	2	2	2	2	0	2	2	0	0	0	0	0	0
Sparsity	0	1	0	1	2	1	1	2	1	1	1	1	0	2
Linearity	1	1	0	0	2	0	2	2	0	0	0	0	0	0
Smoothness	1	2	1	0	2	2	2	2	0	0	0	0	0	0
Monotonicity	1	0	1	0	1	0	0	2	0	0	1	1	0	0
Visualizability	2	2	2	0	2	0	2	2	2	0	0	0	0	0
Sum	**7**	**8**	**6**	**3**	**11**	**3**	**9**	**12**	**3**	**1**	**2**	**2**	**0**	**2**

involved. In Online Appendix F, we provide additional details and an exemplary model output.

Model assessment based on interpretability criteria Based on the detailed model inspections above, it is possible to evaluate all models regarding the various interpretability criteria described in Sect. 3.4. The results of our evaluation process are summarized in Table 6.

As expected, the linear model (i.e., LR) receives the highest interpretability score (12 points) because all criteria are fully satisfied by design.[11] In contrast, the five black-box models receive the lowest scores, with the neural network (i.e., MLP) not fulfilling any of the criteria (0 points) and TabNet and the tree ensemble models (i.e., RF, XGB, CatBoost) having at least the option of incorporating sparsity and/or monotonicity constraints through certain hyperparameters (1 and 2 point(s), respectively).

The scores obtained by the different GAMs are highly scattered. GAMI-Net and IGANN achieve the best scores (11 and 9 points, respectively), since they satisfy most of the criteria by default or at least have additional hyperparameters to address them. NAM and ExNN, on the other hand, are ranked at the lower end of the spectrum with 3 points each. The main deficiencies are the inability to capture linear relationships, missing monotonicity constraints, and a lack of visualizability to clearly understand how a change in the input affects the output, which can also be seen in the examples in Online Appendix F. The remaining three GAMs (i.e., P-Splines, TP-Splines, and EBM) receive medium scores of 7, 8, and 6 points, respectively. This is due to partially fulfilled interpretability criteria or the need to set additional configurations. Thus, depending on the model, suitable hyperparameters must be set in order to meet the corresponding requirements for sparsity, linearity, smoothness, and/or monotonicity (see Online Appendix G).

A final comment should be made on the result of the decision tree. The model is usually considered highly interpretable, but receives low scores in our evaluation procedure. This is due to the fact that shallow decision trees can usually be converted into easily understandable decision rules – an ability that is not appropriately reflected in the current assessment framework adapted from Sudjianto and Zhang (2021). We discuss this limitation in Sect. 5.

4.3 Summary of the Performance-Interpretability Evaluation

In this section, we summarize the results of our previous considerations in order to merge the different evaluation

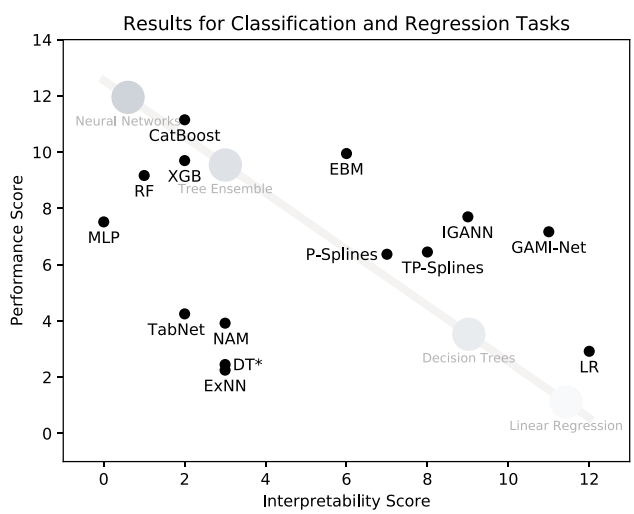

Fig. 5 Summary of the performance-interpretability evaluation. The x-axis represents the interpretability score from the interpretability assessment. The y-axis represents the performance score, computed by inverting the average rank of each model

perspectives into one single view. Specifically, we plot the results of the average performance evaluation based on all tuned models on the *y*-axis, and the results of the interpretability evaluation on the *x*-axis, resulting in the performance-interpretability diagram in Fig. 5.

Such performance-interpretability diagrams have been widely used in recent years to emphasize the trade-off between models with high performance but low interpretability (e.g., neural networks, tree ensembles) versus models with high interpretability but low performance (e.g., linear regression, decision trees) (e.g., Barredo Arrieta et al. 2020; Gunning and Aha 2019). This assumed trade-off, highlighted in gray in Fig. 5, has often been used as an argument in academia and industry to justify the need for complex black-box models. Thus, model developers and researchers have argued that the only way to achieve high predictive performance is the use of complex black-box models, which subsequently require post-hoc explanation methods like SHAP or LIME to simplify their decision logic in a human-digestible way (Rudin 2019; Rudin and Radin 2019). However, this trend is problematic since post-hoc explanation methods cannot represent the full, highly complex functioning of black-box models and therefore may provide unreliable approximations or may lead to misleading conclusions (Rudin and Radin 2019; Babic et al. 2021).

The results of our evaluation study contribute important insights to this central debate. In particular, they challenge the conventional belief and show that there is no strict trade-off between predictive performance and model interpretability for tabular data. Instead, we see that advanced models like EBM, IGANN, and GAMI-Net are indeed able to achieve high predictive performance while remaining fully interpretable without the need for additional explanation methods. On average, EBM even surpasses the performance of most complex black-box models, including XGB, RF, and MLP, providing excellent prediction qualities balanced with decent interpretability. Concurrently, GAMI-Net and IGANN demonstrate excellent interpretability scores while maintaining competitive performance metrics. These relations also hold true if the results for regression and classification tasks are considered separately (see Online Appendix H).

5 Discussion of the Results

In recent years, we have seen a tremendous amount of ML applications across a wide range of application domains, with a primary focus on advanced black-box models due to their expected performance benefits. To mitigate the limitation of such model's opaque behavior, the ML community has brought forth various post-hoc explanation methods to turn the model's complex decision logic into human-understandable insights. For this reason, post-hoc explanation methods are now widely used in a broad variety of disciplines, which can also be observed in our IS community (e.g., Wanner et al. 2020; Bauer et al. 2021; Brasse et al. 2023; Zacharias et al. 2022; Bauer et al. 2023).

However, although post-hoc explanation methods such as SHAP and LIME can offer informative insights, they have to be regarded with caution for two critical reasons. First, the complexity of black-box models can never be fully explained by simple approximations without losing information. As such, they rather provide snapshots of the models' inner decision logic based on a few instances, which may yield incomplete insights. Second, post-hoc explanations can only be provided *post-hoc*, making it impossible to fully validate the functioning of the model for all inputs before model deployment. This becomes particularly critical if the distribution of the input data shifts over time and the model potentially handles value ranges of input features that have not been observed in the training data. As a result, post-hoc explanations have led to misleading conclusions (Rudin and Radin 2019; Babic et al. 2021). Because of these reasons, the option of choosing a complex black-box model that requires post-hoc explanations should only be considered if there is no better alternative (Rudin 2019).

With our study, we show that black-box models are indeed not necessarily required in the first place to achieve strong prediction qualities for tabular data. By conducting a comprehensive and systematic evaluation study, we shed light on the benefits and limitations of advanced intrinsically interpretable ML models. Specifically, this included seven state-of-the-art GAMs, each built upon diverse foundations such as splines, tree-based algorithms, and tailored neural networks. The results of our experiments clearly show that models like EBM, TP-Splines, GAMI-Net, and IGANN do not have to hide behind their opaque counterparts CatBoost, XGB, RF, MLP, and TabNet for tabular prediction tasks. Therefore, we argue that such interpretable models should be firmly established as first-choice models in predictive modeling projects as we see a great potential for their application in research and industry alike.

5.1 Implications for Research and Practice

Our research has several implications for IS research and practice. First and foremost, we hope that our results will promote the use of intrinsically interpretable models in our community as a socio-technical discipline. Because of their flexibility in capturing complex patterns and their simplicity in producing easily understandable results, they

offer a technically equivalent but ethically more acceptable alternative to black-box models. For example, we have seen many examples in the past where bias and fairness issues in ML applications have been reported. Prominent examples include hiring applications that have discriminated against women, or criminal justice applications that have made unfair pretrial detention and release decisions for African Americans due to biased predictions (Janiesch et al. 2021; Rudin and Radin 2019). While detecting bias in training data is generally considered a tedious task, with advanced GAMs such biases can be better detected using the shape plots of the final model to avoid various types of discrimination. For example, using the shape plot of the *crimes* dataset in Fig. 4 as an example, one can quickly see the extent to which the different GAMs capture racial bias effects that may need to be corrected in order to provide fair decision support.

Likewise, model users can better interact with such interpretable ML models because they can directly see how individual features affect the outcome of a model. Thus, domain knowledge can be incorporated into the model by removing or adding features, adjusting certain model properties (e.g., smoothness), or adding other model constraints (e.g., upper and lower bounds on shape functions) to better account for desired patterns. Because of all these characteristics, GAMs can be considered promising models for critical applications in research and practice.

Furthermore, we see great potential in using such models as a tool for theory development. The inherent interpretability of the models facilitates the identification of meaningful predictors, the assessment of their effect, and the generation of hypotheses about causal relationships. Thus, although the observed feature effects are still based on correlations, making it impossible to say with certainty why some effects appear in the feature shape plots, they can still serve as useful indicators for further investigation. Therefore, GAMs can serve as a bridge between data-driven exploration and theory-building in our socio-technical field, helping researchers to uncover hidden phenomena and refine theoretical frameworks.

In addition to providing important insights and implications for our community, our study also provides a versatile evaluation pipeline for systematically evaluating and comparing a variety of ML models. This pipeline encapsulates all of our models, datasets, metrics, and evaluation procedures, thus promoting both reproducibility and reusability of our experiments. The novelty of our pipeline lies in its broad applicability, which is not limited to the models we studied. Instead, it is tailored to be easily adapted to a range of regression or classification datasets for benchmarking a variety of models, including GAMs, traditional interpretable models, and black-box models. Moreover, it allows for the inclusion of custom or proprietary models, which are either implemented in Python or R, to meet individual needs in research and practice.

Ultimately, our study provides a broad and lightweight introduction to GAMs as an important family of interpretable ML models. This introduction shows that interpretable machine learning is not just a research area, but a serious, applicable domain. Therefore, we hope that our work will attract the attention of policy and decision makers in order to promote the relevance of interpretable machine learning beyond the research community. In this way, they might reflect on the responsible and ethical use of AI so that it can guide decisions about the regulation of AI and ML technologies in a reasonable way.

5.2 Limitations and Outlook

As with any research, our work is not without limitations. By acknowledging them, we derive a basis for promising research avenues and future work in this area.

First, we limited our evaluation study to medium-sized datasets to keep the computational cost manageable. This was necessary because we could see that especially the GAMs based on deep neural networks require high computational resources with long training times, leading to a total training time of more than 279 h (see Online Appendix D). However, in future work it might be worthwhile to investigate datasets with different characteristics (e.g., large collections with several million samples and many more features) in order to provide more evidence for our findings.

Second, we explicitly focused on prediction tasks with tabular data, which typically contain naturally meaningful features for interpretation purposes. In domains with higher-dimensional data, such as images and text, the results of this study may not be directly applicable. For this purpose, some upstream feature engineering methods are required that transform raw input data such as pixels or text snippets into higher-level features before feeding them into GAMs to produce interpretable shape functions.

Third, we considered only a limited set of hyperparameters and refrained from using a fully exhaustive tuning grid, which was computationally infeasible. Thus, it is conceivable that both the interpretable as well as the black-box models may achieve even higher predictive performance for certain configurations and datasets. However, since we took care to vary influential parameters in all models and followed the recommendations of the respective model developers and authors, we are confident that our results based on 68,500 training runs provide a fair and representative reflection of the true predictive qualities.

Fourth, the selection of interpretable models in our study was explicitly limited to the family of GAMs. This was

necessary to define a standardized evaluation procedure and to ensure the comparability of the results. However, we would like to emphasize that there are even more types of intrinsically interpretable models, such as Bayesian models and rule-based learners (Barredo Arrieta et al. 2020), but also more advanced (hybrid) approaches, such as logit leaf models (De Caigny et al. 2018) and spline-rule ensembles (De Bock and De Caigny 2021). Extending our systematic model evaluation to such model families would be a promising direction to obtain an even more comprehensive picture of how the individual models are positioned within the machine learning landscape. Furthermore, it would be interesting to use ensembles of different GAM variants (De Bock et al. 2010) to investigate how their predictive performance can be further improved while investigating the impact on interpretability aspects.

A final limitation concerns the evaluation of the models' degree of interpretability. For this purpose, we adopted a qualitative assessment framework based on six evaluation criteria, and scored each model by three experienced developers of this study in three consecutive cycles. This type of "internal" evaluation was necessary because it required several weeks of engagement with each individual model in order to fully grasp their particularities. As a result, it was possible to thoroughly examine and discuss the merits and limitations of each model, and to report our findings in a transparent and reproducible manner. At the same time, however, an external perspective is currently lacking that considers how external model users would assess the level of interpretability of the different models under realistic circumstances (Doshi-Velez and Kim 2017). To this end, it is planned to conduct case studies and field experiments with data science experts and decision makers from different domains and to evaluate the usefulness of the different models addressing real prediction problems.

In addition, we plan to expand the evaluation criteria of the qualitative evaluation framework, as it may not currently provide a fully representative picture of all necessary dimensions of interpretability. For example, it was noticeable that the decision tree received a low interpretability score even though it is commonly classified as a highly interpretable model. At this point, other dimensions are conceivable, such as the convertibility of the model into easily understandable decision rules. Likewise, our evaluation framework currently assumes that all six interpretability criteria are equally important. However, depending on the specifics of the application domain, individual criteria may have a much higher weight, such as the requirement for sparsity in use cases where hundreds of features are involved (e.g., manufacturing, e-commerce) or the requirement for monotonicity in specific applications such as credit scoring (see Online Appendix H). Future work should therefore focus on the extension and context-specific weighting of interpretability criteria for different contexts and application domains.

Furthermore, we believe that it is worthwhile to invest more effort in the identification and examination of GAM-specific interpretability metrics (e.g., number of visual chunks, degree of smoothness, number of discontinuities), so that a more fine-grained interpretability assessment of different GAM variants is possible. Such metrics could also be combined with user-centric evaluation studies to find out which factors have the greatest impact on comprehensibility and usability.

6 Conclusion

The expanding influence of machine learning across various domains makes the adoption of intrinsically interpretable models mandatory, which is particularly true for high-stakes sectors such as healthcare, finance, and judiciary. Our research aimed to evaluate and compare modern variants of such intrinsically interpretable models, namely generalized additive models, with traditional interpretable as well as black-box models to dispel the myth that only black-box models offer strong prediction qualities.

Our systematic study involved an extensive analysis of seven GAMs and seven commonly used ML models based on a large collection of twenty benchmark datasets and an extensive hyperparameter search. Our analysis revealed that advanced GAMs provide a promising alternative to black-box models, offering competitive predictive performance and inherent interpretability. Among the GAMs we evaluated, EBM emerged as the leading model. It outperformed both conventional interpretable models, such as LR and DT, as well as most of the commonly used black-box models, including XGB, RF, and MLP. This finding highlights the potential of GAMs as a preferable choice for high-stakes decision scenarios, in which transparency plays a pivotal role.

Despite these promising results, we acknowledge the limitations of our study, including the focus on medium-sized tabular datasets, a specific set of ML models, and a manageable choice of hyperparameters. In addition, the assessment of interpretability was primarily based on a qualitative evaluation framework that may not capture all dimensions of interpretability. Other criteria may be relevant in different contexts or for different stakeholders. Therefore, future research should address these limitations to provide more nuanced perspectives on this highly relevant topic. Nevertheless, we are confident that our results provide important insights for the IS community, paving the way for a deeper understanding of the predictive performance, interpretability, and applicability of intrinsically interpretable models.

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